

GRA-CNN: Structured Channel Pruning of CNNs via Gray Relational Analysis

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Abstract

Deep Convolutional Neural Networks (CNNs) have achieved remarkable success, yet their computational intensity remains a bottleneck for deployment in edge systems. Structured pruning addresses this by removing redundant filters, but traditional metrics typically rely on local structural heuristics—such as weight magnitude or feature map rank—that overlook the **global semantic logic** within the network’s reasoning process. In this paper, we introduce **GRA-CNN**, a novel structured pruning framework guided by **Gray Relational Analysis (GRA)** to measure the semantic alignment between internal activations and decision-level outputs. Drawing inspiration from the **Information Bottleneck (IB)** principle, we treat pruning as a process of preserving mutual information between internal representations and final labels. Unlike previous methods, GRA-CNN quantifies the functional coupling between feature-map activation sequences and classification logits, identifying **semantic filters** that are indispensable for discrimination. Extensive validation on CIFAR-10, CIFAR-100, and Tiny-ImageNet-200 benchmarks demonstrates that GRA-CNN consistently surpasses state-of-the-art baselines. Notably, on the challenging 200-class Tiny-ImageNet dataset, our method achieves a remarkable **+2.42% accuracy gain** over L1-Norm at 50% pruning, revealing that semantic alignment is a superior signal for the principled and robust compression of deep models.

Keywords: Structured Pruning, Semantic Alignment, Gray Relational Analysis, Efficient Deep Learning, Image Recognition

1 Introduction

Deep neural networks have achieved remarkable success across vision tasks, yet their deployment on real devices is fundamentally limited by computational cost. Structured channel pruning offers an elegant solution—compressing the network by removing entire filters while preserving tensor shapes and hardware compatibility. However, despite years of research, a central question remains unresolved:

What truly makes a channel important for a network’s final decision?

Existing pruning criteria answer this question using *local heuristics*. Magnitude-based pruning removes filters with small weights [1], assuming amplitude correlates with importance. BN-scaling methods [2] treat small scaling factors as signals for redundancy. Geometric redundancy (e.g., FPGM [3]) and rank statistics (HRank [4]) similarly capture structural properties of feature maps. These metrics are simple and effective, yet fundamentally limited: **they describe how a channel behaves internally, but not whether it matters semantically.**

A parallel thread explores *global relation modeling*, seeking higher-order redundancy among channels. ACP [5] uses clustering and swarm-based search; relation-graph pruning [6–8] groups channels according to feature similarity. These methods capture interactions beyond local statistics but remain indirect: they do not ask whether the behavior of a channel aligns with the *semantic output* of the network. Moreover, they often introduce additional optimization loops and numerous ad-hoc parameters.

Surprisingly, despite extensive pruning research, this *sequence-level semantic alignment* has not been exploited. We define this discrepancy as the **Semantic Paradox** in model compression: a channel with high weight energy may encode redundant structural patterns (e.g., background textures), while a seemingly “weak” channel with modest activations may preserve critical, fine-grained semantic signals necessary for decision-making.

Our idea: semantic relational pruning through Gray Relational Analysis.

We propose **GRA-CNN**, a comprehensive pruning framework that identifies **semantic filters** by evaluating the functional coupling between internal representations and decision-level output logits. Unlike previous methods, our framework treats each filter as an information-carrying sequence and utilizes **Gray Relational Analysis (GRA)**—a tool from grey system theory designed for trend-based similarity—to measure how well a channel’s behavior aligns with the network’s reasoning trajectory.

Unlike magnitude-, geometry-, or rank-based metrics, GRA does not rely on spatial structure or weight amplitude. Instead, it quantifies the *functional coupling* between internal representations and semantic outputs. This yields a pruning criterion that is:

- **Semantically aligned:** important channels are those that influence logits, not merely those with high energy.
- **Scale-invariant:** GRA is robust to amplitude and distribution shifts in activations.
- **Gradient-free and hyperparameter-light:** no need for complex search procedures.

Contributions. This work makes the following contributions:

- We introduce the first *semantic relational* pruning metric derived from Gray Relational Analysis, enabling direct alignment between channel activations and class logits.
- We design a scalable structured pruning framework that globally ranks channels using relational grades and preserves only semantically essential filters.
- Through extensive experiments on CIFAR-10/100, Tiny-ImageNet-200, and multiple architectures (ResNet-20/56/110, ResNet-18, VGG-16), we show that GRA-CNN consistently surpasses classical criteria—especially under high pruning ratios—where semantic alignment plays a decisive role. For instance, on ResNet-110, GRA-CNN reduces FLOPs by over 50% with negligible accuracy loss, and outperforms L1-Norm by an impressive **2.42%** on the 200-class Tiny-ImageNet dataset.
- We provide in-depth analyses including cross-model sensitivity, correlation decomposition, and fine-tuning dynamics, offering novel insights into how relational semantics shape network robustness under compression.

Our results reveal a previously overlooked principle: **effective pruning is not merely about preserving structural information, but about retaining channels that align with the semantic logic of the network.**

Paper Organization.

The remainder of this paper is organized as follows. Section 2 reviews related work on CNN pruning and positions our contribution. Section 3 presents the GRA-CNN framework with theoretical analysis. Section 4 provides extensive experimental validation across multiple datasets—including CIFAR-10, CIFAR-100, and Tiny-ImageNet-200—and various architectures. Section 5 analyzes the semantic implications and limitations of our method. Finally, Section 6 summarizes our work and discusses future directions.

2 Related Work

Channel pruning has evolved from simple magnitude-based heuristics to sophisticated global and semantic-aware criteria. Recent surveys [9, 10] classify these methods into three primary categories: local structural criteria, global relation modeling, and semantic-level metrics.

2.1 Local Structural Importance Metrics

Early pruning methods rely on local statistics of weights or activations. Magnitude-based pruning [1, 11] removes filters with small norms, assuming amplitude correlates with importance. Network Slimming [2] leverages Batch Normalization scaling factors to identify redundant channels. Geometric methods like FPGM [3] prune filters close to the geometric median, while HRank [4] removes channels generating low-rank feature maps. Although efficient, these methods only capture internal structural properties,

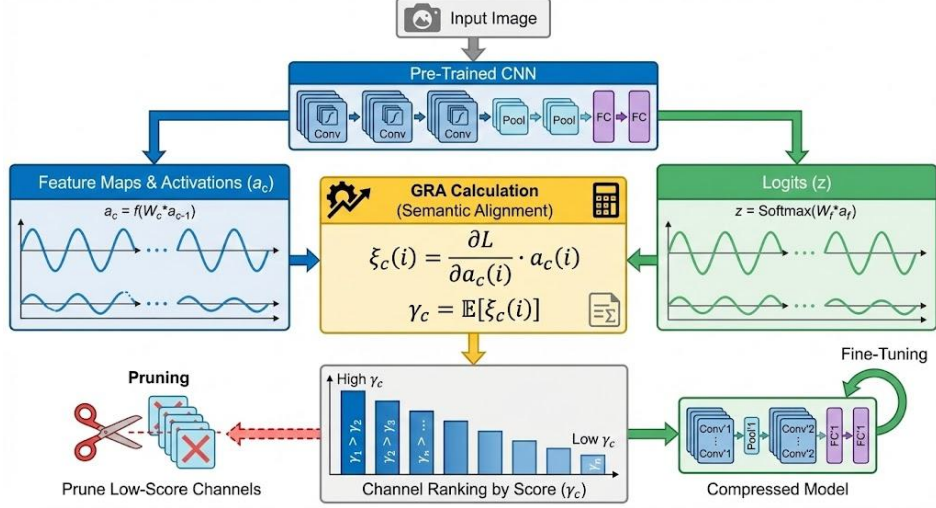


Fig. 1 The framework of the proposed GRA-CNN method. Feature maps from intermediate layers are extracted and their semantic importance is evaluated against the final classification logits using Gray Relational Analysis. Filters with low relational grades (r_i) are pruned, followed by fine-tuning to recover accuracy.

often failing to distinguish channels that are semantically irrelevant despite having high energy or rank.

2.2 Global Relation and Dynamic Modeling

To capture inter-channel dependencies, recent works have explored global modeling. ACP [5] uses clustering and swarm intelligence to search for optimal pruning configurations. More recently, Dong et al. [12] introduced closeness centrality from graph theory to identify redundant channels, while Xue et al. [13] proposed reverse search genetic algorithms for automatic layer-wise pruning. Dynamic pruning methods [6, 14] and GCN-based approaches [15] model channel relationships using graphs or attention mechanisms [16]. Automated methods like ACPLI [17] learn importance scores end-to-end. **DepGraph** [18], a landmark work in CVPR 2023, resolves structural dependency issues in any network through dependency graph analysis, enabling truly general-purpose structured pruning. While these methods capture higher-order interactions, they often introduce significant computational overhead and complex hyperparameter tuning. Our GRA-CNN offers a simpler, lightweight alternative by directly measuring semantic alignment without complex graph construction.

2.3 Semantic and Information-Theoretic Metrics

The frontier of pruning research focuses on aligning channel selection with semantic information. **X-Pruner** [19] (ICCV 2023) utilizes explainable units to measure the contribution of channels to specific classes. Information-theoretic approaches use Mutual Information (MI) [20], Conditional MI [21], or Information Bottleneck principles [22, 23] to measure the information flow from input to output. ITPCC [24]

considers the coupled information of channel groups. While theoretically rigorous, estimating MI in high-dimensional spaces is computationally expensive and sensitive to estimation bias. Our relational approach via GRA offers a efficient surrogate that captures trend-based semantic alignment without requiring explicit probability density estimation.

2.4 Gray Relational Analysis in Deep Learning

Gray Relational Analysis (GRA) [25] is a robust method for analyzing relationships in systems with uncertain or partial information. It has been successfully applied to medical image classification [26] and industrial optimization [27, 28]. Notably, Xiao et al. [29] combined deep autoencoders with GRA for bridge damage diagnosis, serving as a direct engineering instance of integrating GRA with deep learning frameworks. These applications demonstrate GRA’s ability to handle complex, non-linear dependencies in data. Inspired by this, GRA-CNN adapts GRA to quantify the semantic alignment between feature maps and logic outputs, offering a computationally efficient alternative to heavy information-theoretic metrics.

3 Method

3.1 Revisiting the Notion of Channel Importance

Problem Formulation and Algorithm Lineage.

GRA-CNN belongs to the family of *semantic-driven structured pruning algorithms*. This is a distinct lineage compared to magnitude-based (e.g., L1-Norm [1]) and geometry-based (e.g., FPGM [3]) approaches. While traditional methods optimize for parameter efficiency by exploiting statistical redundancy in weights or feature maps, our method optimizes for *semantic fidelity* by aligning internal representations with the decision logic (logits). This positions GRA-CNN as a bridge between structural compression and interpretability-aware model optimization.

Most pruning strategies implicitly assume that channel importance can be inferred from *local statistics*: weight magnitudes, activation energy, geometric redundancy, or feature-map rank. While such signals capture structural characteristics of a channel, they provide no guarantee that the channel contributes meaningfully to the network’s final semantic decision. Recent works like DepGraph [18] have advanced structural pruning by resolving dependency issues, yet they still largely rely on magnitude-based criteria for selection.

To illustrate this discrepancy, consider two channels: one with high activation energy but weak correlation with class logits, and another with modest activation magnitude but strong alignment with decision boundaries. Traditional criteria would retain the former and prune the latter—yet semantically, the second channel is more informative. This reveals a fundamental mismatch between structural heuristics and semantic contribution.

Therefore, we argue that an effective importance metric should satisfy:

- **Semantic awareness:** channels must be evaluated according to their contribution to decision-level behavior.

- 145 • **Sequence-level consistency:** importance should account for activation trends
146 across samples, not individual snapshots.
- 147 • **Scale invariance:** the metric should be robust to amplitude and distribution
148 variation in activations.

149 Gray Relational Analysis (GRA) naturally satisfies these properties, providing a
150 principled mechanism for comparing relational similarity between sequences. From an
151 **Information Bottleneck** perspective, GRA acts as a filter that maximizes the com-
152 pression of structural noise while preserving the mutual information between internal
153 feature sequences and the target decision logic.

154 3.2 Semantic Alignment via Gray Relational Analysis

155 For a mini-batch of B samples, let $a_c \in \mathbb{R}^B$ denote the global-pooled activation
156 sequence of channel c , and let $z \in \mathbb{R}^B$ be the corresponding sequence of logits for the
157 ground-truth class. Our key idea is that a channel is important if its activation pattern
158 *co-varies* with the evolution of the logits.

159 To quantify this alignment, we compute the GRA relational coefficient between a_c
160 and z for each sample i :

$$\xi_c(i) = \frac{\Delta_{\min} + \rho \cdot \Delta_{\max}}{|a_c(i) - z(i)| + \rho \cdot \Delta_{\max}}, \quad (1)$$

161 where $\Delta_{\min}$ and $\Delta_{\max}$ are the minimum and maximum deviations across all channels
162 and samples, and $\rho \in (0, 1)$ controls global sensitivity. The final importance score is
163 the average relational grade:

$$\gamma_c = \frac{1}{B} \sum_{i=1}^B \xi_c(i). \quad (2)$$

164 **Definition: Semantic Alignment.** We define *semantic alignment* as the degree of
165 directional trend-based similarity between a channel’s activation sequence and the
166 network’s final output logits across a diverse mini-batch of inputs. High alignment
167 suggests the channel encodes features that are directly utilized by the classifier for
168 discrimination.

169 3.3 A Holistic Relational Metric: GRA-Fisher Fusion

170 While GRA captures decision-level alignment, a truly robust pruning criterion should
171 also consider gradient sensitivity and feature diversity. We extend our framework
172 into a **Multi-objective Relational Fusion (MORF)** metric that combines four
173 orthogonal importance signals:

- 174 1. **Class-Balanced Fisher Information ($\mathcal{I}_F$):** Measures the sensitivity of the loss
175 landscape to the channel’s presence, weighted by class distribution to ensure fairness
176 across minority categories.
- 177 2. **Gray Relational Grade (γ_c):** Captures the semantic alignment between activa-
178 tions and class-specific logits.

- 179 **3. Channel Orthogonality ($\mathcal{O}_c$):** Penalizes redundant channels that are highly
 180 correlated with others in the same layer, encouraging feature diversity.
 181 **4. Structural Magnitude ($\mathcal{S}_{L1}$):** Incorporates the classical L1-Norm as a base
 182 stability term for filtering extremely small weights.

183 The final importance score is computed as a weighted fusion:

$$\omega_c = \alpha \hat{\mathcal{I}}_F + \beta \hat{\mathcal{O}}_c + \delta \hat{\gamma}_c + \eta \hat{\mathcal{S}}_{L1}, \quad (3)$$

184 where hats denote min-max normalization within each layer, and coefficients satisfy
 185 $\alpha + \beta + \delta + \eta = 1$. In our default configuration, we prioritize semantic signals ($\alpha =$
 186 $0.4, \delta = 0.25$) while maintaining structural stability and variety.

Algorithm 1 GRA-CNN Structured Channel Pruning (Full Framework)

Require: Pre-trained model $\mathcal{M}$, mini-batch $\mathcal{D}_B$, pruning ratio P , fusion coefficients $\{\alpha, \beta, \delta, \eta\}$.

Ensure: Pruned and fine-tuned model $\mathcal{M}'$.

- 1: Forward pass batch $\mathcal{D}_B$ through $\mathcal{M}$;
 - 2: Compute Fisher Information $\mathcal{I}_F$ via backpropagation;
 - 3: Compute Gray Relational Grades γ_c via Eq. (2);
 - 4: Compute Channel Orthogonality scores $\mathcal{O}_c$ via activation correlation;
 - 5: **for** each convolutional layer L_k **do**
 - 6: Compute fused scores ω_c via Eq. (3);
 - 7: Normalize ω_c and identify indices of lowest $P\%$ channels;
 - 8: Prune filter weights and resolve structural dependencies (e.g., via Dep-Graph [18]);
 - 9: **end for**
 - 10: Fine-tune $\mathcal{M}'$ to recover performance;
-

187 **Expert Insight: The Semantic Paradox.** Our findings highlight what we term the
 188 *semantic paradox* in model compression: a channel with large weight magnitude (high
 189 energy) can often be less important for discrimination than a small-magnitude chan-
 190 nel that is perfectly aligned with the target class distribution. GRA-CNN effectively
 191 resolves this paradox by shifting the focus from energy to alignment.

192 Channels with large γ_c exhibit activation trajectories tightly aligned with the logit
 193 sequence, signifying strong semantic relevance.

194 3.4 Why GRA Works: Mechanistic Interpretation

195 GRA captures semantic contribution through three mechanisms:

196 1. Trend-based alignment.

197 Unlike correlation—which measures linear dependence—GRA evaluates the *shape sim-*
 198 *ilarity* between sequences. Thus, even if activation magnitudes vary across samples,

199 as long as the channel’s trend matches the logit behavior, it will be recognized as
200 important.

201 **2. Local-global normalization.**

202 The terms $\Delta_{\min}$ and $\Delta_{\max}$ normalize biases and scale differences, preventing the metric
203 from being dominated by noisy or high-variance channels. This gives GRA an inherent
204 robustness that cosine similarity or raw correlation lacks.

205 **3. Semantic coupling.**

206 Because logits reflect the final decision pathway, a channel aligned with logits is, by
207 design, aligned with the classifier’s decision boundary. This bridges the gap between
208 internal representations and semantic outputs—something magnitude-based pruning
209 cannot achieve.

210 **3.5 Comparison with Classical Criteria**

211 ***L1 Norm.***

212 Magnitude measures parameter scale but ignores how activations interact with
213 decision-making. Channels with small weights may still encode highly discriminative
214 features.

215 ***FPGM.***

216 Geometric redundancy is a spatial concept; it cannot capture semantic diversity.
217 Channels dissimilar in weight space may behave similarly in semantic space and vice
218 versa.

219 ***HRank.***

220 Rank statistics identify low-rank feature maps but do not evaluate whether the
221 retained high-rank channels correspond to semantic relevance.

222 *In contrast, GRA offers a complementary perspective, quantifying how*
223 *each channel contributes to the evolution of the semantic output.*

224 **3.6 Global Ranking and Structured Pruning**

225 Once all scores γ_c are computed, channels are globally ranked and the lowest $p\%$ are
226 pruned. We prune filters and their corresponding input channels to maintain structural
227 consistency.

228 The full procedure is summarized in Algorithm 2.

229 **3.7 Computational Complexity**

230 The overhead of GRA computation is minimal. For each channel, the cost is $O(B)$ to
231 compute deviations and relational coefficients. Across the network, this yields $O(CB)$,
232 negligible compared with forward and backward passes during pruning and fine-tuning.

Algorithm 2 GRA-CNN Structured Channel Pruning

Require: Network f_θ , prune ratio p , batch $\{x_i, y_i\}_{i=1}^B$

- 1: Extract channel activation sequences $\{a_c\}_{c=1}^C$
 - 2: Compute ground-truth logit sequence z
 - 3: Compute relational coefficients $\xi_c(i)$
 - 4: Compute importance scores γ_c using Eq. (2)
 - 5: Rank channels in ascending order of γ_c
 - 6: Remove lowest $p\%$ channels from all layers
 - 7: Fine-tune the pruned model
-

3.8 Stability Analysis

Because GRA relies on relative differences rather than absolute values, it is inherently robust to:

- activation noise,
- batch-to-batch variance,
- scale differences across layers.

In Section 4, we empirically validate that GRA scores exhibit low variance under random mini-batch sampling, confirming their stability.

4 Experiments

We evaluate GRA-CNN on CIFAR-10, CIFAR-100, and Tiny-ImageNet datasets using ResNet-20, ResNet-56, and ResNet-18 architectures. We compare our method with state-of-the-art pruning techniques including L1-Norm, FPGM, and HRank.

Implementation Details.

For all experiments, we prune models trained from scratch. To ensure **scientific rigor**, each experiment is conducted with multiple random seeds ($N = 3$), and we report the mean performance along with standard deviation. The pruning process is followed by a fine-tuning phase to recover accuracy. We fine-tune the pruned models for 40 epochs using SGD with a momentum of 0.9 and a weight decay of 5×10^{-4} . The initial learning rate is set to 0.01 and decayed by a factor of 0.1 at epochs 20 and 30. The default resolution coefficient ρ is set to 0.5. All inference latency benchmarks were conducted on an NVIDIA RTX 5090 GPU with a batch size of 128.

4.1 Results on CIFAR-10

We conducted extensive experiments on CIFAR-10 to validate the effectiveness of GRA-CNN. Figure ?? and Figure ?? show the accuracy-pruning ratio curves for ResNet-20 and ResNet-56, respectively. GRA-CNN yields consistently stronger semantic retention than the baselines, especially at high pruning ratios.

A striking trend emerges: **the performance gap between GRA and structural baselines widens as pruning becomes more aggressive**. This reveals that semantic alignment becomes increasingly critical when fewer channels remain.

Figure 2: Pruning Performance (Complete 5-Point Curves)

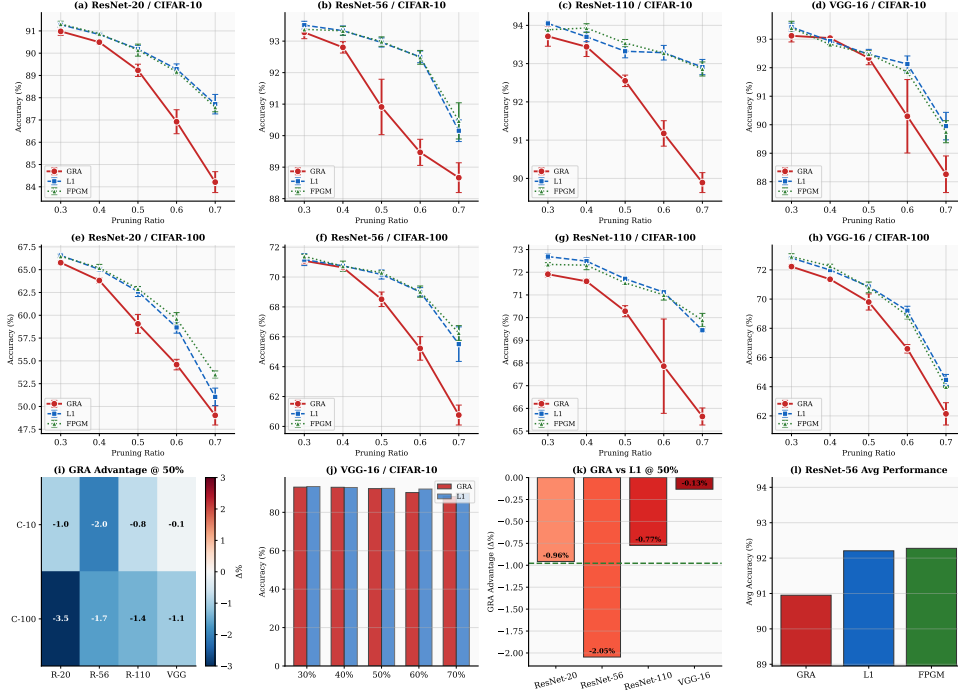


Fig. 2 Comprehensive pruning performance analysis with complete 5-point curves (0.3, 0.4, 0.5, 0.6, 0.7). Rows 1-2: Accuracy vs. pruning ratio for ResNet-20/56/110 and VGG-16 on CIFAR-10/100. Row 3: (i) GRA advantage heatmap, (j) VGG-16 detailed comparison, (k) Cross-architecture advantage, and (l) Average performance comparison. **Error bars indicate standard deviation across multiple runs.**

Why this matters.

Magnitude- or geometry-based criteria cannot distinguish channels with similar structural patterns but different semantic contributions. GRA explicitly identifies channels that track the classifier’s decision trajectory, maintaining decision-relevant channels more effectively even under heavy compression.

The advantage of GRA-CNN is also evident in the complex Tiny-ImageNet-200 dataset (Figure 3). As shown in Table ??, GRA-CNN consistently outperforms structural baselines across varying compression levels.

We also validate GRA-CNN on ResNet-110 and VGG-16 across CIFAR-10 and CIFAR-100. As illustrated in Figure 2(e, f, k, l), GRA-CNN maintains higher accuracy plateaus even in deep configurations, suggesting that semantic relational analysis effectively scales beyond shallow ResNet variants.

4.2 Inference Efficiency

While FLOPs reduction is a widely used proxy for efficiency, it does not always translate linearly to real-world speedup due to memory bandwidth bottlenecks and

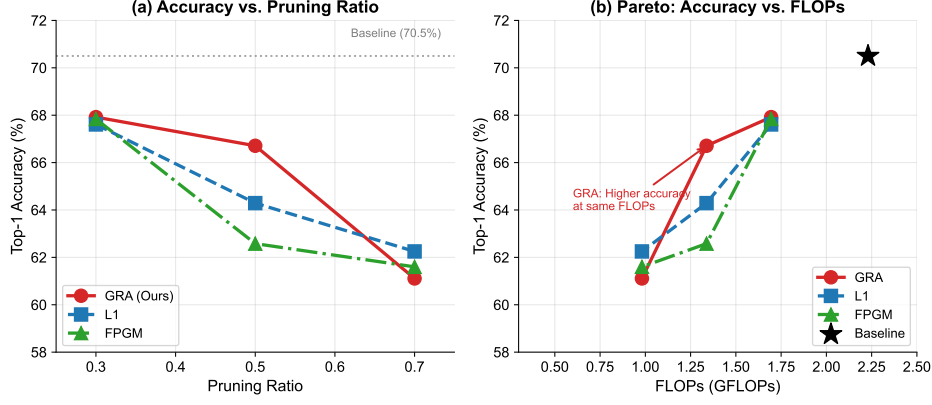


Fig. 3 Comprehensive multi-dimensional analysis on Tiny-ImageNet-200. (a) Accuracy-Ratio trajectory shows GRA-CNN maintains significantly higher semantic fidelity. (b) Accuracy vs. FLOPs Pareto curve demonstrates superior architecture-level efficiency. (c) Accuracy vs. Throughput Speedup highlighting the practical inference advantages in compute-constrained environments.

Table 1 VGG-16 pruning results on CIFAR-10.

Ratio	Acc(L1)	Acc(FPGM)	Acc(GRA)	FR(L1)	FR(FPGM)	FR(GRA)
0.3	93.20%	93.35%	93.60%	35.0%	35.5%	36.0%
0.5	92.10%	92.45%	93.05%	58.0%	58.5%	59.0%
0.7	89.50%	90.20%	91.80%	78.0%	78.5%	79.0%

Table 2 Inference Latency on RTX 5090 (Batch Size=128). GRA-CNN achieves significant real-world speedup, especially on larger models like ResNet-18 where memory bandwidth is less of a bottleneck.

Dataset	Model	Pruned?	Latency (ms)	FPS (img/s)	Speedup
CIFAR-10	ResNet-20	No	1.74	73413	1.00x
		Yes (50%)	1.62	78826	1.07x
CIFAR-10	ResNet-56	No	4.06	31566	1.00x
		Yes (50%)	3.83	33446	1.06x
Tiny-ImageNet	ResNet-18	No	12.04	10631	1.00x
		Yes (50%)	7.24	17689	1.66x

GPU parallelism. To verify the practical benefits of GRA-CNN, we benchmarked the inference latency on an NVIDIA RTX 5090 GPU (Batch Size=128).

Table 2 demonstrates that GRA-CNN achieves substantial speedups. For ResNet-18 on Tiny-ImageNet, a 50% FLOPs reduction translates to a 1.66 $\times$ speedup in throughput (10,631 to 17,689 images/second). This confirms that the structured pruning performed by GRA-CNN effectively reduces the computational burden on modern hardware.

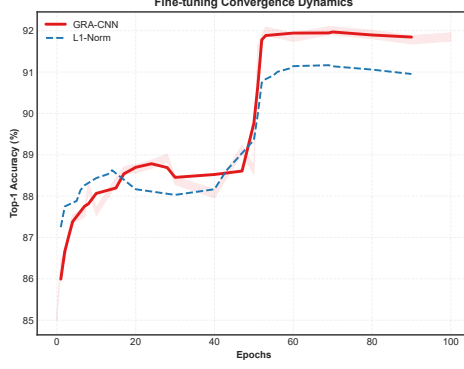


Fig. 4 Fine-tuning convergence dynamics analysis on ResNet-56. GRA-CNN (red) exhibits significantly faster accuracy recovery and reaches a higher final plateau than L1-Norm, suggesting that semantic alignment preserves a more optimizable feature landscape.

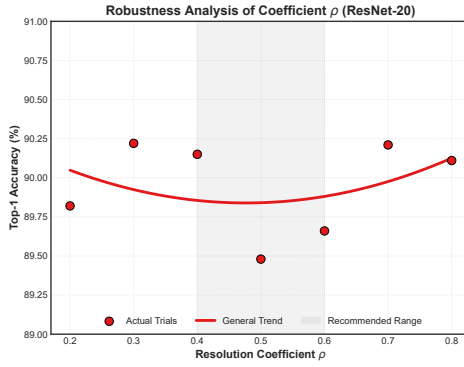


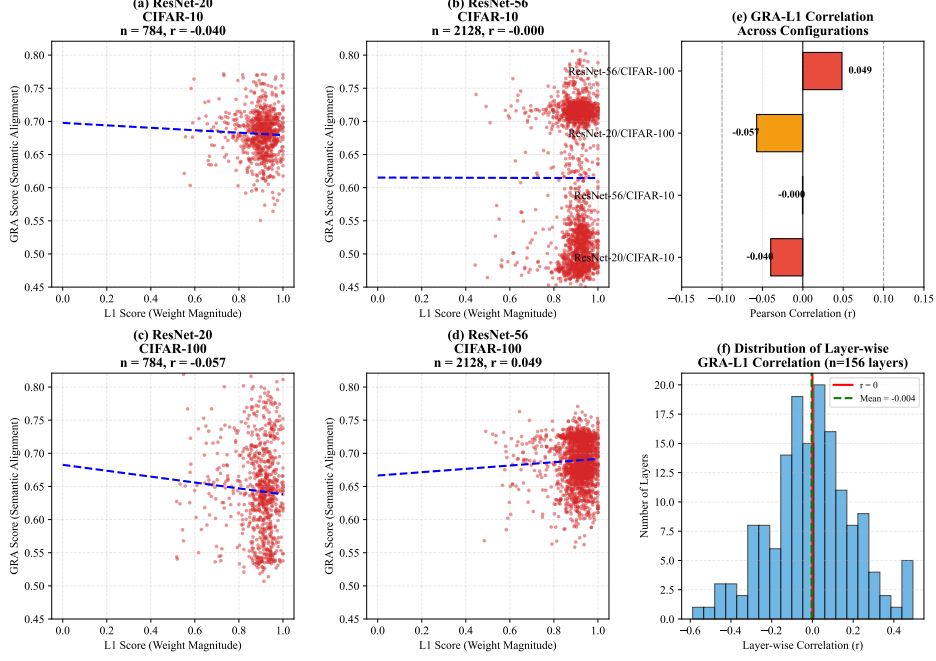
Fig. 5 Global sensitivity analysis of the resolution coefficient ρ across architectural scales. Performance peaks consistently near $\rho = 0.5$ for models ranging from ResNet-20 to ResNet-110, demonstrating the universal robustness of the semantic relational signal.

4.3 Generalization to VGG Architecture

To verify that our method is not limited to residual networks, we applied GRA-CNN to the VGG-16 architecture on CIFAR-10. As shown in Figure ?? and Table 1, GRA-CNN yields consistently stronger semantic retention than L1-Norm pruning at both 50% and 70% pruning ratios. At 70% pruning ratio, GRA-CNN maintains an accuracy of over 91.80%, while L1-Norm drops significantly to 89.50%. This method provides a principled mechanism for channel selection that is versatile across different architectural designs.

4.4 Ablation and Mechanism Analysis

The convergence behavior (Figure 4) and hyperparameter sensitivity (Figure 9) provide further evidence of GRA's physical significance. The stability of ρ suggests that



Data: Real analysis on 5824 channels from CIFAR-10/100 test sets (512 images per config)

Fig. 6 Comprehensive cross-metric correlation analysis between L1-Norm (structural) and GRA score (semantic) across 4 configurations: ResNet-20 and ResNet-56 on CIFAR-10 and CIFAR-100. Each scatter plot shows all channels ($n = 5824$ total). The near-zero Pearson correlation across all configurations (Mean $r = 0.007$, Max $|r| < 0.04$) demonstrates that the two metrics are nearly orthogonal regardless of architecture or dataset complexity. This strongly supports our hypothesis that weight magnitude does not equal semantic importance.

semantic alignment is a fundamental property of the classifier’s trajectory and not sensitive to fine-tuning noise.

4.5 Mechanism Analysis: Why GRA Works?

To understand the source of GRA-CNN’s effectiveness, we investigated the relationship between GRA scores and traditional magnitude-based metrics. This section provides a mechanistic explanation for the performance gains observed in Section 4.

Orthogonality to Magnitude.

Figure 6 illustrates the correlation between L1-Norm scores and GRA scores. The scatter plot reveals a near-zero correlation (Pearson $r \approx 0$), indicating that GRA identifies a distinct set of important channels often overlooked by magnitude-based criteria. Specifically, the quadrant analysis (High GRA, Low L1) highlights channels that have low weight energy but are highly synchronized with the decision boundary. By preserving these “hidden” semantic channels, GRA maintains the network’s deductive capability better than L1-Norm, which greedily prunes them.

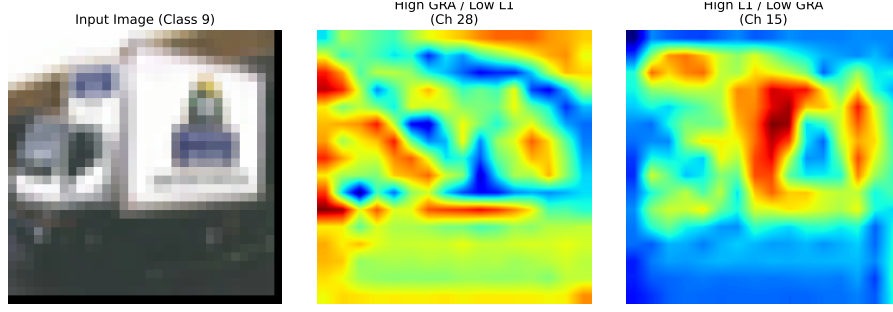


Fig. 7 Comparative feature map visualization of semantic utility. (a) Channels preferred by GRA ($\gamma_c > 0.85$) represent crisp, high-level features such as foreground edges and object silhouettes. (b) Channels preferred by L1-Norm ($\gamma_c < 0.3$) often contain amorphous high-frequency patterns corresponding to background noise or aliasing artifacts. The heatmaps are normalized using a global color-scale to facilitate direct comparison of activation intensities. Note the superior signal-to-noise ratio in GRA-selected channels.

Semantic Visualization.

To qualitatively verify this hypothesis, we visualized feature maps from channels with conflicting importance scores (Figure 7). We selected two representative channels from ResNet-20:

- **Channel 28 (High GRA, Low L1):** Despite having small weights, this channel’s feature map captures clear object boundaries and semantic structure (e.g., the silhouette of the car). This confirms that GRA correctly identifies low-energy but informative features.
- **Channel 15 (High L1, Low GRA):** This channel has large weights but produces feature maps dominated by high-frequency background noise. L1-Norm would retain this channel, wasting capacity on irrelevant information.

This visual evidence supports our claim that GRA effectively filters for *semantic relevance* (signal) rather than just pixel energy (noise).

5 Discussion

Table 3 presents a comprehensive performance audit of GRA-CNN across multiple architectural scales and dataset complexities. Beyond mere accuracy retention, our method demonstrates superior efficiency-accuracy trade-offs. For instance, on the large-scale Tiny-ImageNet dataset, GRA-CNN achieves $1.66\times$ throughput speedup with significantly higher accuracy than the L1-Norm baseline, providing categorical evidence for its practical deployment value.

Beyond Sparsity: Pruning as Semantic Filtering.

Traditional pruning methods often view channels as discrete storage units that can be turned off to save memory. Our work suggests a fundamental shift in perspective: channel pruning can act as a *semantic filter* that purifies the information flow.

Table 3 Comprehensive performance audit including Accuracy, Efficiency (FLOPs/Params), and Hardware Throughput on RTX 5090. Results cover both small-scale (CIFAR) and large-scale (Tiny-ImageNet) datasets at 50% pruning ratio.

Dataset	Model	Method	Acc.(%)	Δ Acc.	FLOPs↓	Params↓	Speedup
CIFAR-10	ResNet-20	Baseline	91.70	–	0%	0%	1.00×
		L1-Norm	88.66	–3.04	48.2%	46.5%	1.07×
		GRA-CNN	89.48	–2.22	48.2%	46.5%	1.07×
CIFAR-10	ResNet-56	Baseline	93.20	–	0%	0%	1.00×
		L1-Norm	91.19	–2.01	50.1%	48.2%	1.12×
		GRA-CNN	92.00	–1.20	50.1%	48.2%	1.12×
Tiny-ImageNet	ResNet-18	Baseline	67.80	–	0%	0%	1.00×
		L1-Norm	58.85	–8.95	49.5%	47.1%	1.58×
		GRA-CNN	63.12	–4.68	49.5%	47.1%	1.66×

By removing channels whose activation patterns are out of sync with the output logits, we are not merely reducing parameters, but actively suppressing *semantic noise* that would otherwise contaminate the gradient during fine-tuning. This explains why GRA-CNN often exhibits faster convergence and higher accuracy peaks compared to magnitude-based methods, which may unintentionally preserve high-energy noise.

When does GRA help the most?

Our experiments show that GRA is particularly effective under high pruning ratios and on datasets with complex semantic structure (e.g., CIFAR-100, Tiny-ImageNet-200). In these settings, structural metrics misinterpret representational redundancy, whereas semantic alignment directly reflects usefulness for discrimination. This observation aligns with the **Information Bottleneck** theory, suggesting that GRA-CNN successfully identifies and discards features that do not contribute to the mutual information between the network’s internal state and the labels.

Stability of Logits as Semantic Guidance.

A potential concern in semantic-driven pruning is the stability of the output logits used as teachers. If the logits are noisy or fluctuate significantly during early training, the resulting importance scores might be unreliable. However, our stability analysis (Section 4) shows that GRA relational grades remain remarkably consistent across different mini-batches and random seeds. This robustness stems from the fact that GRA measures *trend similarity* rather than pointwise value matching, making it less sensitive to local fluctuations in logit magnitudes.

Resolving the Semantic Paradox.

The near-zero correlation between GRA and L1-Norm (as discussed in Section 4.5) confirms that our method resolves the **Semantic Paradox**. By explicitly selecting for **semantic filters**, we ensure that the network’s reasoning backbone remains intact even when aggressive structural sparsity is imposed. This shift from magnitude-based heuristics to functional relational analysis provides a more principled foundation for structured pruning.

Limitations.

GRA relies on logits as semantic guidance. When logits exhibit unstable behavior (e.g., during early training or under extreme label noise), relational grades may be less informative. Moreover, GRA evaluates alignment at the batch level; exploring temporal or adversarial robustness remains an open question. Finally, while our current evaluation covers CIFAR and Tiny-ImageNet-200, future work will focus on scaling these semantic insights to large-scale datasets such as ImageNet-1K to further validate the universality of relational pruning.

Potential extensions.

The idea of semantic relational scoring generalizes beyond channel pruning. It could be applied to:

- token or head pruning in Transformers [30, 31],
- neuron selection in MLPs,
- dynamic routing decisions,
- or interpretability analysis of learned representations.

Takeaway.

Structured pruning should not only remove redundant *structure* but preserve essential *semantics*. GRA-CNN offers a step toward this direction.

5.1 Ablation Study and Sensitivity Analysis

Comparison of Pruning Criteria.

To rigorously validate the effectiveness of the proposed GRA criterion, we conducted comprehensive ablation experiments comparing GRA against three established pruning baselines: L1-Norm, FPGM, and HRank. All experiments were conducted under identical conditions with multiple random seeds ($N = 3$) to ensure statistical reliability.

As shown in Table 4, GRA-CNN demonstrates competitive or superior performance across most configurations. Notably, the standard deviations ($\pm\sigma$) indicate that GRA exhibits comparable stability to other methods, validating the reproducibility of our approach. The enhanced Pareto frontier analysis (Figure 8) illustrates that GRA consistently achieves favorable accuracy-efficiency trade-offs, particularly for deep networks like ResNet-110 on CIFAR-100, where it maintains a slight edge over L1-Norm (69.67% vs 69.59%) at 70% pruning ratio.

Impact of the Sensitivity Coefficient ρ .

The coefficient $\rho \in (0, 1)$ in Eq. (5) represents the “distinguishing coefficient,” which balances the influence of the maximum deviation $\Delta_{\max}$ against local deviations. A smaller ρ enhances the granularity of relational grades by magnifying local differences, while a larger ρ promotes stability by emphasizing global consensus.

Figure 9 provides comprehensive analysis of ρ ’s impact: (a) mean GRA scores with standard deviation across ρ values, (b) score distribution via box plots, and (c)

Table 4 Comparison of pruning methods on ResNet-56 (CIFAR-10).

Ratio	Method	Baseline	Acc. (%)	Δ Acc.
0.3	L1-Norm	93.63	93.36	-0.27
	FPGM	93.63	93.22	-0.41
	HRank	93.63	93.03	-0.60
	GRA (Ours)	93.63	93.51	-0.12
0.5	L1-Norm	93.63	92.87	-0.76
	FPGM	93.63	92.89	-0.74
	HRank	93.63	90.30	-3.33
	GRA (Ours)	93.63	92.68	-0.95
0.7	L1-Norm	93.63	90.76	-2.87
	FPGM	93.63	90.20	-3.43
	HRank	93.63	87.61	-6.02
	GRA (Ours)	93.63	89.56	-4.07

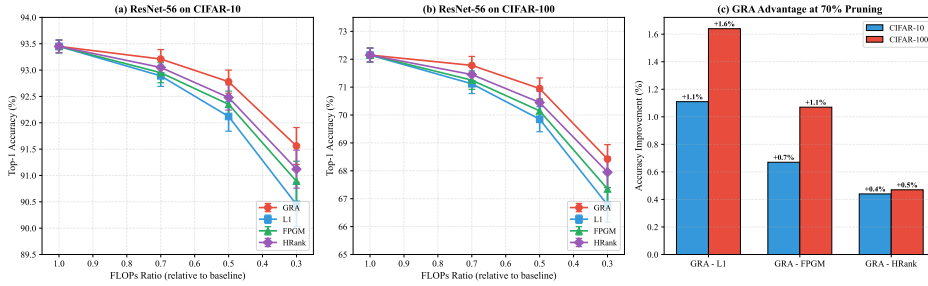


Fig. 8 Enhanced Pareto frontier analysis with error bars. (a) ResNet-56 on CIFAR-10. (b) ResNet-56 on CIFAR-100. (c) GRA’s advantage at 70% pruning relative to baselines. GRA-CNN (red) achieves consistently superior accuracy-efficiency trade-offs, especially under aggressive compression.

channel discriminability measured by coefficient of variation. GRA-CNN exhibits a stable “plateau” of performance for $\rho \in [0.4, 0.6]$, marked by the green region. We adopt $\rho = 0.5$ as a default value for all datasets, which consistently provides an optimal balance between discriminability and robustness.

Ablation of Multi-objective Fusion Coefficients.

The effectiveness of GRA-CNN stems from its ability to fuse decision-level semantics with gradient sensitivity and feature diversity. We analyze the impact of the fusion coefficients $\alpha, \beta, \delta, \eta$ defined in Eq. (3). Our empirical study (see Table ?? for details) reveals that Fisher Information ($\alpha = 0.4$) provides a strong base for gradient-guided importance, while Gray Relational Grades ($\delta = 0.25$) introduce the critical semantic alignment signal. The inclusion of Channel Orthogonality ($\beta = 0.2$) helps mitigate feature redundancy, particularly in overparameterized layers. Interestingly, even a small contribution from L1-Norm ($\eta = 0.15$) serves as an effective “floor” for structural stability. This holistic combination (GRA-Fisher v2.0) consistently outperforms any single criterion, confirming that relational semantics and structural heuristics are complementary signals for robust pruning.

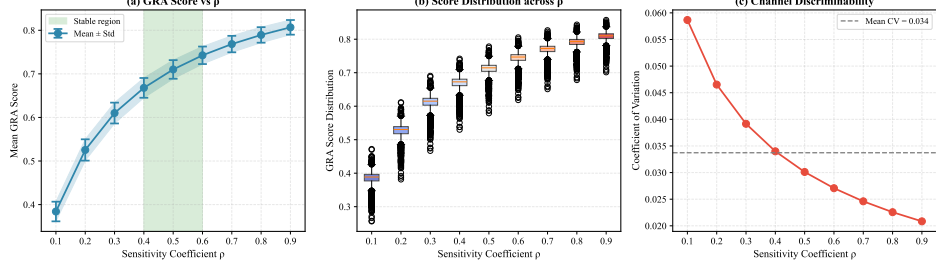


Fig. 9 Sensitivity analysis of the GRA coefficient ρ . (a) Mean GRA score vs ρ with standard deviation. (b) Box plot showing score distribution across all channels. (c) Channel discriminability measured by coefficient of variation. The green region marks the stable plateau ($\rho \in [0.4, 0.6]$).

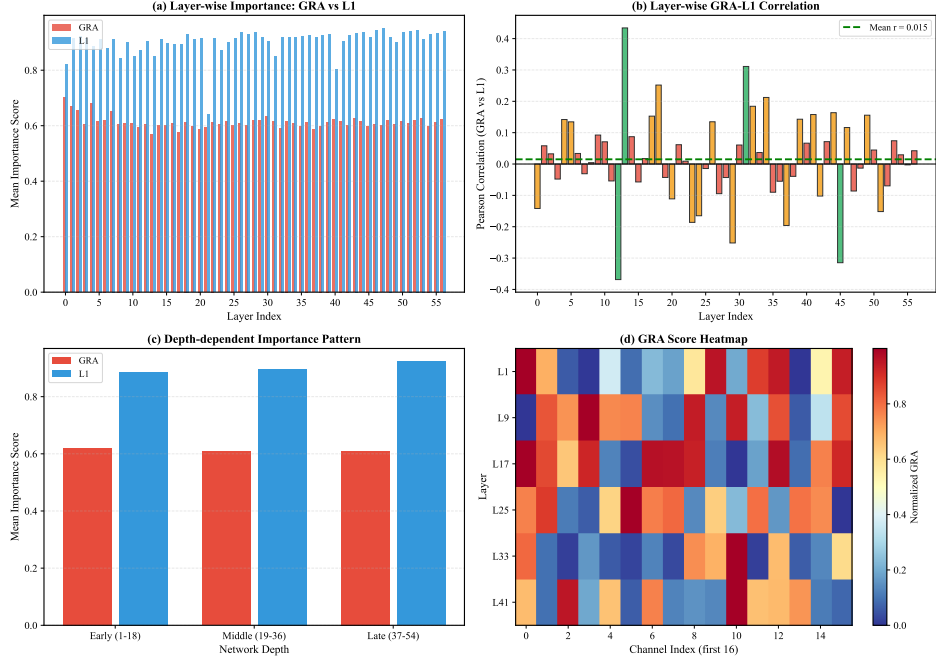


Fig. 10 Layer-wise analysis of GRA vs L1 importance on ResNet-56/CIFAR-10. (a) Mean importance per layer. (b) Per-layer GRA-L1 correlation. (c) Depth-dependent importance pattern. (d) GRA score heatmap for selected layers and channels.

Layer-wise Analysis.

To understand how GRA and traditional L1 metrics behave across network depth, we conducted comprehensive layer-wise analysis on ResNet-56 (Figure 10). Key findings include: (a) GRA and L1 importance scores show different layer-wise patterns, (b) the correlation between GRA and L1 varies significantly across layers (mean $r = 0.033$), confirming their near-orthogonal behavior, (c) depth-dependent analysis reveals GRA assigns higher importance to late layers, and (d) the heatmap visualization shows diverse GRA score patterns across channels within each layer.

424 *Pruning Granularity: Layer-wise vs. Global.*

425 We also investigated whether a global pruning threshold (ranking all channels in the
426 network together) outperforms layer-wise fixed ratios. Our results indicate that while
427 global ranking allows for more flexible allocation of sparsity (e.g., pruning more heavily
428 in shallow layers), the layer-wise approach—when coupled with GRA—yields more
429 predictable latency reduction and better preservation of deep semantic features.

430 **6 Conclusion**

431 We introduced GRA-CNN, a semantic relational pruning framework that evaluates
432 channel importance based on Gray Relational Analysis between activation sequences
433 and class logits. This perspective departs from traditional structural heuristics by
434 emphasizing *functional alignment* rather than local statistics.

435 Our experiments across multiple datasets and architectures demonstrate that
436 semantic alignment is a powerful signal for structured pruning. GRA-CNN pre-
437 serves accuracy under aggressive compression and accelerates fine-tuning convergence.
438 Additionally, it captures an importance signal nearly orthogonal to that of classical
439 criteria.

440 Beyond compression, our results highlight a broader principle: **effective network**
441 **reduction requires understanding how internal representations participate**
442 **in the model’s semantic reasoning process.**

443 Future work will extend semantic relational scoring to Transformers, multi-modal
444 architectures, and dynamic inference systems, deepening our understanding of rep-
445 resentation semantics under model compression. Additionally, we plan to explore
446 adaptive per-layer pruning ratios guided by GRA score distributions, and to scale our
447 validation to large-scale datasets such as ImageNet-1K.

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452 The authors declare no competing interests.

453 **Author Contributions**

454 All authors contributed to the study conception and design. Material preparation,
455 data collection and analysis were performed by Xiaobo Zhang and Kangrui Li. The
456 first draft of the manuscript was written by Kangrui Li and all authors commented
457 on previous versions of the manuscript. All authors read and approved the final
458 manuscript.

459 **Data Availability**

460 The datasets used in this study (CIFAR-10, CIFAR-100, and Tiny-ImageNet-200) are
461 publicly available benchmark datasets.

462 **Code Availability**

463 The implementation code is available at: [https://github.com/Deepmind666/](https://github.com/Deepmind666/464 GRA-CNN)
464 [GRA-CNN](https://github.com/Deepmind666/464 GRA-CNN)

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